Question 2: *MCQs* To be answered in **(18) Minutes** [ ] / 6 Marks

YOU MUST USE THE PROVIDED TABLE BELOW. USE CAPITAL LETTERS (A, B, C, D)

1	2	3	4	5	6

1. Which of the following Big O notations represents the best (most efficient) time complexity?
   A. $O(n^2)$
   B. $O(lg(n!))$
   C. $O(2^{lg\, n})$
   D. $O(n\, lg\, n)$

2. Consider the recurrence: $T(n)= 4T(n/5) +5n$, using the Master method, choose the correct answer.
   A. Apply case 1 and $T(n) = \Theta(n^{\log_5 4})$
   B. Apply case 2 and $T(n) = \Theta(n^{\log_5 4} \log n)$
   C. Apply case 3 and $T(n) = \Theta(n)$
   D. Master method cannot be used.

3. Consider a binary max-heap implemented using an array. Which one of the following arrays represents a binary max-heap?
   A. 30, 13, 10, 2, 3, 14, 11
   B. 30, 13, 10, 14, 11, 2, 3
   C. 30, 13, 14, 2, 3, 10, 11
   D. 30, 3, 14, 10, 11, 2, 13

4. Which of the following is not a stable sorting algorithm in its typical implementation?
   A. Counting sort
   B. Merge sort
   C. Insertion sort
   D. Heap sort

5. Suppose we are sorting an array of eight integers using heapsort, and we have just finished some heapify (either maxheapify or minheapify) operations. The array now looks like this: 16, 14, 15, 10, 12, 27, 28. How many heapify operations have been performed on root of heap?
   A. 1
   B. 2
   C. 3
   D. 4

6. What is the worst-case time complexity of insertion sort where position of the data to be inserted is calculated using binary search?
   A. $O(n^2)$
   B. $O(n))$
   C. $O(n^2\, log\, n)$
   D. $O(n\, log\, n)$

Question 1: ***Short Answers*** To be answered in **(18) Minutes** [ ] / **6 Marks**

1. Show that $f(n) = 3n^2 + 2n + 4$ *is* $\Theta(n^2)$ and find the constants $c1$, $c2$ and n0? **[2 marks]**

2. Indicate for each pair of expressions (A, B) in the table below, whether A is O, o, Ω, ω, or θ of B. Your answer should be in the form of table with 'yes' or 'no' written in each box. **[4 marks]**

A	B	O	o	Ω	ω	θ
2^n	$2^{n/2}$					
$n^{\log m}$	$m^{\log n}$					
$\log n + 100$	$\log n^2 + n^3$					
$n^{(7/4)}$	$n \log_2(n)$					

Question 3: ***Recurrence*** To be answered in **(25) Minutes** [ ] / **8 Marks**

A. Given the following recurrence:

$$T(n) = \begin{cases} 1, & n = 1 \\ T\left(\frac{n}{2}\right) + n^2, & n > 1 \end{cases}$$

1. Draw the recursion tree based on the recurrence. **[2 marks]**

2. Determine the number of levels in the tree. **[1 mark]**

3. Calculate the total cost (*Big-O*) of *T(n)* of all levels. **[2 mark]**

B. Solve the recurrences below using Master method **[3 marks]**

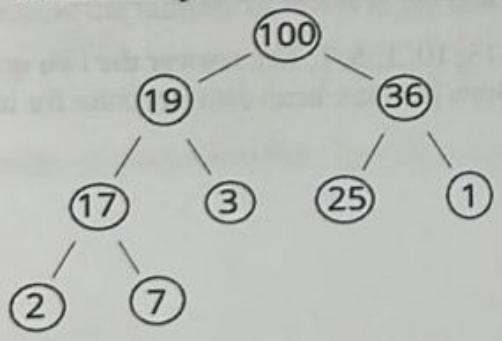
1. $T(n) = 2T(n/2) + n \log n$

2. $T(n) = 8\,T(n/2) + n^3$

Student Name (in English): _____ Student ID: _____

2. If necessary, apply MAX-HEAPIFY function to the max-heap data structure in (1) to ensure that all nodes satisfy the max-heap property. Show your work. **[1 mark]**

C. Given the following max-heap, show the resultant heap after the first iteration of HEAPSORT(A), i.e., draw the max-heap after the first iteration of HEAPSORT(A). **[1 mark]**



Question 4: *Sorting* To be answered in **(15) Minutes** [ ] / 5 Marks

A. Consider the following array:

4	2	3	0	9	1	4

1. Consider using counting sort, write the auxiliary array C' (counting array) after accumulating the results. **[1 mark]**

C'

0	1	2	3	4	5	6	7	8	9

2. Write the output array B after the first iteration **[1 mark]**

B

B. Given the following array A [23, 17, 14, 6, 13, 10, 1, 5, 7, 12], answer the two questions below:
   1. Is the array A a max-heap (Yes/No)? Draw the max-heap data structure for array A. **[1 mark]**